

Predicting local control of choroidal melanomas following ^{106}Ru plaque brachytherapy

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ABSTRACT

Background We evaluated the control rate of choroidal melanomas treated with ^{106}Ru plaque brachytherapy to identify the risk factors associated with local recurrence and lack of response.

Methods A retrospective review of ^{106}Ru plaque brachytherapy for patients with choroidal melanoma treated at St Bartholomew's Hospital, London. Survival analysis was used to assess associations between evaluated age, sex, location, foveal proximity, tumour base and height, presence of lipofuscin and subretinal fluid, apex dose, radiation rate and type of plaque with time to local recurrence. Logistic regression analysis was used to assess to evaluate the association between the same set of variables and lack of tumour response.

Results From January 2002 to December 2006 189 patients were treated. The follow-up ranged from 12 to 78 (median 33) months. None of the patients received adjuvant diode laser thermotherapy. The control rate was 85.7% (14 recurred while 13 did not respond). Of the patients who had local recurrence, univariate survival analysis demonstrated an association with younger patients, foveal proximity, preoperative subfoveal fluid and tumour base >11 mm. Age and foveal proximity remained significant in a Cox multiple variable model ($p=0.03$). Of the patients who did not respond, logistic regression analysis showed that lack of response was associated with a tumour height >5 mm, confirmed through multiple variable analysis ($p=0.027$).

Conclusions Tumours that are close to the fovea in young patients appear more likely to show local recurrence. Tumour height >5 mm was the only prognostic factor that determined lack of response. These results may be used to select which tumours require adjuvant therapy.

INTRODUCTION

Published data on the growth rate of uveal melanoma and effect on tumour control of different treatment modalities have raised important concerns over the appropriate choice of treatment.^{1–7} Since the introduction of ^{106}Ru plaque brachytherapy by Lommatzsch and Vollmar in 1964,⁸ this type of globe-conserving procedure has been widely accepted in many centres throughout Europe.^{9–11}

In 1991 the Collaborative Ocular Melanoma Study demonstrated that for medium-sized choroidal melanoma, patient survival was comparable with enucleation.¹² Plaque brachytherapy offers the patient a perceived better quality of life with the potential to preserve vision and the eye.¹³ However, the application of ^{106}Ru plaque brachytherapy is limited by a steep dose-gradient in tissue, with the upper limit of penetration judged to be

approximately 5 mm.¹⁴ Therefore, this radioisotope is only suitable for small to medium-sized melanomas. Adjuvant treatment with diode laser thermotherapy may allow better tumour control for more elevated melanomas with the potential to reduced radiation complications.^{15–19}

During the study period, however, diode laser thermotherapy was not used as routine adjuvant therapy for uveal melanoma and was only performed for local recurrence.

The aim of this study was to identify the clinical factors associated with local tumour recurrence and lack of tumour response following treatment of choroidal melanoma with ^{106}Ru plaque brachytherapy in order to provide an evidence-based strategy to select patients who may benefit from adjuvant diode laser thermotherapy.

PATIENTS AND METHODS

A retrospective review was undertaken on patients with posterior uveal melanomas, treated with ^{106}Ru plaque brachytherapy from January 2002 to December 2006, at the Ocular Oncology Unit of St Bartholomew's Hospital in London.

The inclusion criteria were patients with posterior uveal melanomas (choroidal tumours with no ciliary body component) treated solely with ^{106}Ru plaque brachytherapy, and who had at least 1 year of follow-up in the service. Patients were excluded from the study if there was insufficient follow-up information in the medical records or if they had received adjuvant diode laser thermotherapy.

The diagnosis of posterior uveal melanoma was based on the findings of comprehensive ophthalmic examination supplemented by B scan ultrasonography and fundal photography. Tumours were treated with ^{106}Ru brachytherapy, using commercial radioactive plaques (BEBIG, Berlin, Germany). Both indirect ophthalmoscopy and transillumination were used to demarcate tumour margins intraoperatively. Dummy plaques were placed initially to ensure that all tumour margins were within the field of irradiation. All tumours received at least a 2 mm safety margin. Active plaques were subsequently positioned and left in place until the prescribed treatment dose had been delivered to the tumour apex. After irradiation, follow-up visits took place at 1, 3, 6 and 12 months post-brachytherapy, then twice a year until the second post-operative year and annually thereafter. Follow-up evaluation included best corrected visual acuity, ophthalmoscopic and ultrasonographic assessment of local tumour control and brachytherapy-related complications.

All data were collected retrospectively. Demographic data included age, sex and race of the

patient. Ocular parameters included laterality (right, left), presence of cataract, presence of co-existing retinal disease such as hypertensive or diabetic retinopathy, or age-related macular degeneration. Tumour characteristics included height, maximal base diameter, tumour location (macular, peripapillary, pre-equatorial and post-equatorial), foveal proximity (distance from the fovea (mm), proximity to the optic nerve (mm)), and the presence of lipofuscin or subretinal and subfoveal fluid. Both foveal and optic nerve distances were either retrieved from the initial assessments as documented in the notes, or when possible calculated from the fundus photograph of the tumour. Radiation parameters included type of ^{106}Ru plaque (20 mm CCB, 20 mm notched COB and 15 mm CCA), prescribed apex dose (Gy) and radiation rate (Gy/h). The follow-up period from the time of treatment to the latest date at which the ocular status was known was noted.

The three outcome measures were tumour control, non-response (tumours who failed to show signs of regression or continued to increase within the first year from treatment) and tumour recurrence (documented expansion of the tumour margin or an increase in tumour height confirmed by sequential ultrasound examination following an initial period of tumour control).

STATISTICAL ANALYSIS

Survival analyses (Cox proportion hazards models) were used to examine associations between the hazard of recurrence and the putative risk factors. Only patients who showed an initial response were included in this analysis. All factors were examined in univariate analyses and then stepwise regression was conducted on all variables. Similarly, logistic regression was used to examine associations between putative risk factors and the odds of non-response. All analyses were conducted using the STATA version 10 statistical software (StataCorp, Texas, USA).

RESULTS

Overall, 214 patients diagnosed with posterior uveal melanoma underwent treatment solely with ^{106}Ru plaque brachytherapy from January 2002 to December 2006. Of these patients, six were excluded as they were advised on enucleation but opted for brachytherapy, 13 patients had inadequate follow-up, five had missing documentation on plaque parameters or tumour size, while one patient underwent enucleation within the first year from treatment. Therefore 189 patients were analysed. The patient demographics, tumour characteristics and radiation parameters are summarised in table 1.

The median follow-up period from the initiation of therapy was 33 months (mean 33.6, SD 15.6, range 12–78 months). The overall control rate was 85.7%. Fourteen of 189 patients recurred and 13 of 189 patients did not respond within the first year from treatment. Time to recurrence ranged between 12 and 71 months with a median of 25.5 months.

Tumour height ranged from 0.8 to 10.3 mm (mean 4, SD 1.7 mm). The median maximal base diameter was 9.5 (IQR 8.4–2.5) mm. Regarding tumour location, 28% of tumours involved the macula, 25% were peripapillary, 41% were pre-equatorial while the 6% of tumours were post-equatorial. Peripapillary tumours were treated with a 20 mm COB-notched plaque. The median radiation rate was 1.4 (IQR 1.08 to 1.76) Gy/h with 65% of choroidal melanomas receiving 110–110 Gy, 19.5% receiving 80 Gy and 15.5% 120 Gy to the tumour apex.

The results of Cox univariate survival analysis on 176 patients (excluding the non-responders from the study group) are

Table 1 Demographics of 189 patients with posterior uveal melanoma treated with ^{106}Ru brachytherapy from January 2002 to December 2006, with characteristics of posterior uveal melanomas and plaque radiotherapy parameters

Variable	Description	Median (IQR)	n (%)
Demographics			
Follow-up (months)		33 (24–46)	
Age (years)		62 (53–71)	
Eye	Right		92 (49)
	Left		97 (51)
Sex	Female		102 (54)
	Male		87 (46)
Medical history	Diabetes mellitus		17 (9)
	Hypertension		63 (33)
	Cataract		24 (13)
Characteristics of posterior uveal melanomas			
Location	Macular		53 (28)
	Pre-equatorial		78 (41)
	Post-equatorial		11 (6)
	Peripapillary		47 (25)
Foveal distance (mm)		3 (0.5–5)	
	<1 (foveal)		47 (32.5)
	>1<4 (macular)		48 (33)
	>4 (extramacular)		50 (34.5)
Optic nerve distance (mm)		4 (1.5–6.5)	
	<2		51 (35)
	>2		94 (65)
Tumour base (mm)		9.5 (8.4–11.4)	
	<8		33 (20)
	8–11		81 (49.5)
Tumour height (mm)		3.7 (2.7–5.2)	
	<3		60 (32)
	3–5		77 (41)
	>5		51 (27)
Subretinal fluid			106 (56)
Subfoveal fluid			30 (16)
Lipofuscin			91 (48)
Plaque radiotherapy			
Plaque shape	20 mm CCB		84 (44.5)
	20 mm notched COB		47 (25)
	15 mm CCA		56 (29.5)
Apex dose (Gy)	80		37 (19.5)
	100–110		122 (65)
	120		29 (15.5)
Radiation rate (Gy/h)		1.4 (0.8–1.8)	
	<1		30 (16)
	1–1.6		94 (50)
	>1.6		29 (34)

summarised in table 2. Univariate evaluation of clinical factors related to the risk of local recurrence revealed an association of decreasing hazard with increasing age (HR per year increase 0.96; 95% CI 0.92 to 0.99; $p=0.019$) and with increasing distance from the fovea (HR 0.71; 95% CI 0.51 to 0.98; $p=0.038$). There was a greater hazard of recurrence in patients with pre-treatment subfoveal fluid than without ($p=0.03$, HR 3.25) and evidence of increasing hazard with increasing basal diameter ($p=0.032$, HR 1.25). There was some suggestion that men had an increased risk of local recurrence, although this was not statistically significant with these data ($p=0.08$). Subsequent Cox stepwise analysis, including all variables (except plaque type shape, which was collinear with location), confirmed the association with age

($p=0.096$, adjusted HR 0.043) and proximity to the fovea ($p=0.04$, adjusted HR 0.069), as seen in table 2.

Univariate logistic regression (table 3) looking at the same set of factors associated with non-response showed evidence of association with tumour height ($p=0.006$, OR 1.55). Successive multiple variable logistic regression (table 3), confirmed that the only factor associated with the likelihood of non-response was the height of the tumour ($p=0.027$, OR 1.52).

Figure 1 shows the temporal relationship of tumour recurrence. Even after 60 months of follow-up, the risk of local recurrence was still present. The Kaplan–Meier curve shows a gradual decline in local control with time, as there appears to be no plateau. Figure 2 demonstrates that tumours more than 4 mm from the fovea had a reduced risk of local recurrence.

DISCUSSION

Over the past 10 years ^{106}Ru has been the sole radioisotope used for plaque brachytherapy in the London ocular oncology service. This study was designed to report the success of local tumour control using ^{106}Ru plaque brachytherapy alone without adjuvant diode laser thermotherapy. The selection of a particular radioisotope may influence local control rate. In 1999, we reported our experience with ^{106}Ru , ^{125}I and proton beam radiotherapy. Five-year melanoma recurrence rates were 11%, 4% and 5%, respectively.²⁰ In this study we report a 5-year melanoma recurrence rate of 14% for ^{106}Ru plaque brachytherapy. This is similar to the 5-year recurrence rate published by Karlsson *et al* for ^{60}Co plaque brachytherapy.²¹ However, Lommatzsch *et al*⁴ reported a higher recurrence rate of 37% at 15 years using ^{106}Ru plaque brachytherapy. This recurrence rate may be reflected by the longer period of follow-up in this paper.

In addition, our study demonstrates that the Kaplan–Meier curve for local tumour control does not plateau within the first 5 years of treatment and therefore long-term follow-up is required in these patients.

Adjuvant diode laser thermotherapy can be used to improve local recurrence rates significantly. Damato *et al*²² reported a 5-year recurrence rate of only 2.1% using ^{106}Ru plaque brachytherapy combined with diode laser when appropriate. Irradiated tumours received adjuvant diode laser thermotherapy to the posterior tumour margin. However, this low rate of recurrence may be related to case selection, as many more tumours are selected for proton beam radiotherapy by this centre. Similarly, Shields and associates¹⁵ were able to reduce the 5-year recurrence rate to 3% by combining ^{125}I plaque brachytherapy with adjuvant diode laser thermotherapy. However, comparing local control rates between different institutions may not be reliable, as there is considerable inter-study variation and case mix differences. An important outcome of this study was the development of the ability to select which tumours were more likely to show local recurrence.

In our series, Cox multivariate analysis of prognostic factors for local recurrence identified a higher risk of recurrence in tumours close to the fovea ($p=0.04$). This increased risk of recurrence for juxtafoveal tumours was not due to plaque placement to spare vision, as during the period of the study none of the macular plaques were decentred. Some juxtafoveal tumours were treated with lower radiation rates in order to preserve vision and this could be an explanation for the high risk of recurrence in this group. However, multivariate analysis demonstrated that the rate of delivery of radiation did not affect the risk of local recurrence. It is most likely that the increased blood supply to this area allows tumour cells to survive even the

Table 2 Cox univariate hazard model on 176 patients undergoing ^{106}Ru brachytherapy (survival analysis excluding the non-responders and evaluation of significant clinical factors associated with time to recurrence) and Cox multiple variable analysis on 176 patients (excluding the non-responders from the study group) with significant clinical factors related to the time of recurrence

Variable	Description	HR	p Value	95% CI
Cox univariate hazard model				
Age	Per year increase	0.96	0.019	0.92 to 0.99
Sex	Female	1	—	—
	Male	2.82	0.081	0.87 to 9
Tumour location	Macular	1	—	—
	Pre-equatorial	0.78	0.75	0.17 to 3.53
	Post-equatorial	1.27	0.84	0.12 to 3.1
	Peripapillary	1.74	0.353	0.47 to 7.83
Foveal distance	Per mm increase	0.71	0.038	0.51 to 0.98
Optic nerve distance	Per mm increase	0.85	0.141	0.68 to 1.05
Tumour base	Per mm increase	1.25	0.032	1.02 to 1.53
Tumour height	Per mm increase	1.26	0.014	0.95 to 1.66
Lipofuscin	Present versus absent	1.33	0.605	0.44 to 3.97
Subretinal fluid	Present versus absent	1.28	0.664	0.41 to 3.91
Subfoveal fluid	Present versus absent	3.25	0.030	1.12 to 9.40
Apex dose (Gy)	80	1	—	—
	100–110	0.72	0.638	0.17 to 2.88
	120	2.29	0.273	0.51 to 10.11
Radiation rate	Per Gy/h increase	0.79	0.673	0.27 to 2.29
Plaque shape	20 mm CCB	1	—	—
	20 mm notched COB	2.03	0.247	0.61 to 6.69
	15 mm CCA	0.03	0.278	0.03 to 2.63
Cox multiple variable analysis				
Age	Per year increase	0.96*	0.03	0.92
Foveal distance	Per mm increase	0.68*	0.03	0.48

*Adjusted HR.

Table 3 Univariate logistic regression analysis on 189 patients undergoing ^{106}Ru brachytherapy with evaluation of clinical factors associated with lack of initial response to treatment, and multiple variable regression analysis with significant clinical factors associated with lack of initial response to treatment

Variable	Description	OR	p Value	95% CI
Univariate logistic regression analysis				
Age	Per year increase	1.01	0.625	0.96 to 1.05
Sex	Female	1	—	—
	Male	1	0.993	0.32 to 3.11
Tumour location	Macular	1	—	—
	Pre-equatorial	0.83	0.801	0.21 to 3.28
	Post-equatorial	1.22 0.83	0.862	0.12 to 12.1
	Peripapillary		0.82	0.17 to 3.94
Foveal distance	Per mm increase	1.18	0.085	0.97 to 1.43
Optic nerve distance	Per mm increase	1.09	0.293	0.92 to 1.31
Tumour base	Per mm increase	1.14	0.26	0.9 to 1.44
Tumour height	Per mm increase	1.55	0.006	0.13 to 2.12
Lipofuscin	Present versus absent	0.91	0.881	0.29 to 2.84
Subretinal fluid	Present versus absent	1.83	0.328	0.54 to 6.17
Subfoveal fluid	Present versus absent	0.96	0.96	0.2 to 4.5
Apex dose (Gy)	80	1	—	—
	100–110	2.19	0.47	0.26 to 18.41
	120	7.5	0.07	0.82 to 68.25
Radiation rate	Per Gy/h increase	1.33	0.578	0.48 to 3.7
Plaque shape	20 mm CCB	1	—	—
	20 mm notched COB	0.65	0.537	0.16 to 0.71
	15 mm CCA	0.35	0.197	2.56 to 1.72
Multiple variable regression analysis				
Height	Per mm increase	1.52	0.027	1.04 to 2.21

face of radiation. It is well established that an effective vasculature supply is an independent prognostic factor for tumour growth.²³ Several clinical trials using colour Doppler imaging in patients with choroidal melanoma have identified an increase in blood flow velocity in untreated tumours.^{24 25}

Surprisingly in this series, proximity to the optic disc did not emerge as a significant predictive factor for local recurrence. Twenty-five per cent of melanomas in this study were peripapillary and therefore notched plaque was successful without diode laser thermotherapy.

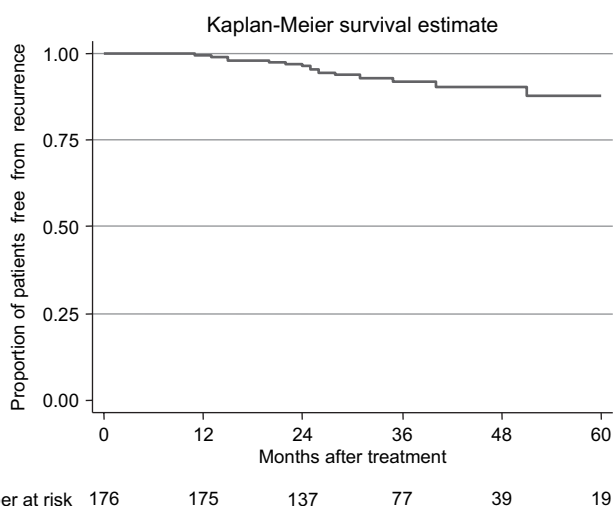


Figure 1 Kaplan–Meier survival estimate on 176 patients with posterior uveal melanoma and showing the temporal relationship of tumour recurrence. Even after 60 months (5 years) of follow-up, the risk of local recurrence was still present. There is a gradual decline in local control with time, but there is no plateau.

Increasing tumour height was identified as a risk factor for lack of response to ^{106}Ru plaque brachytherapy. Tumour height is a predictive factor for local relapse as well as for loss of visual acuity.^{17 26 27} In our series the median tumour height was 3.7 mm. However, 27% of tumours had a thickness of greater than 5 mm and the median tumour height in the non-response group was 5.6 mm, while in the recurrence group it was 4.9 mm. In our retrospective series reported in 1999, treatment with ^{106}Ru plaque brachytherapy was found to be associated with a higher rate of local recurrence than either the ^{125}I or proton beam radiation therapy.²⁰ However, fewer side effects and better visual results were seen in the ^{106}Ru plaque brachytherapy-treated patients. Therefore the use of ^{106}Ru was continued, although we attempted to improve the local tumour control rate

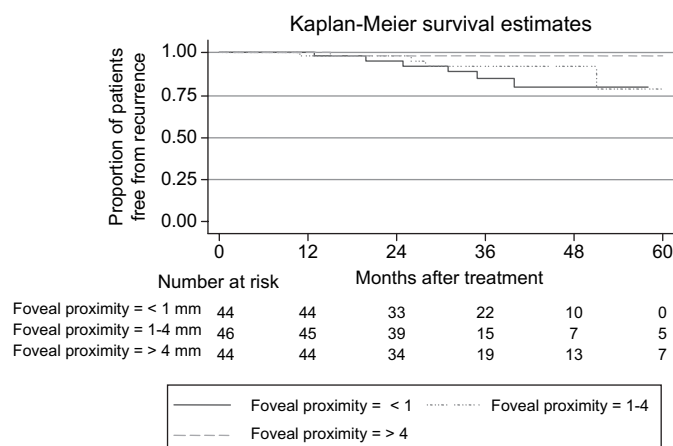


Figure 2 Kaplan–Meier estimate on 176 patients in relation to foveal distance. Cox multivariate analysis showed increased risk of recurrence with foveal proximity ($p=0.03$).

by both using wider treatment margins (through a larger 20 mm plaque) and by varying the radiation dose according to tumour thickness (120 Gy calculated at 5 mm depth).

The upper limit of ^{106}Ru plaque brachytherapy is assumed to be a tumour thickness of 5 mm as β radiation is limited by the steep dose-gradient in tissue.¹⁴ This limitation can be overcome by adopting 'sandwich therapy', where plaque brachytherapy is combined with up to three sessions of adjuvant diode laser thermotherapy.²⁸ Several centres have reported improved control rates with this treatment approach.^{16 17}

There are several limitations that should be realised in this retrospective series, the main one being the follow-up interval for local recurrence and the risk of bias arising from the retrospective collection of data. We have demonstrated that recurrence may occur more than 5 years after treatment. The data regarding lack of initial tumour response and its association with increasing tumour height is therefore most valid. We have identified a reduced 5-year local control rate within our institution for ^{106}Ru plaque brachytherapy: 14% in 2009 versus 11% in 1999 (statistically significant difference, Fisher's exact=0.404, one-sided Fisher's exact=0.215). This may have occurred because thicker tumours were selected for ^{106}Ru plaque brachytherapy since ^{125}I plaques were abandoned. An important observation from this study is that ^{106}Ru brachytherapy as a sole modality of treatment should not be used for tumours exceeding the 5 mm in height. Currently only patients with a tumour height of less than 5 mm are selected for ^{106}Ru plaque brachytherapy.²⁹ Plaque brachytherapy is combined with adjuvant diode laser thermotherapy for tumours greater than 5 mm and up to 7 mm in height. Diode laser thermotherapy may also be considered for juxtafoveal tumours although this approach is likely to result in early visual loss or a paracentral scotoma.

Competing interests None.

Ethics approval The study was approved by the St Bartholomew's Hospital Clinical Governance Department and the Research Government Committee of Moorfields Eye Hospital (Reference no. PAPK100).

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REFERENCES

1. Eskelin S, Pyrhönen S, Summanen P, *et al.* Tumor doubling times in metastatic malignant melanoma of the uvea: tumor progression before and after treatment. *Ophthalmology* 2000;**107**:1443–9.
2. Char DH, Kroll S, Phillips TL. Uveal melanoma. Growth rate and prognosis. *Arch Ophthalmol* 1997;**115**:1014–18.
3. Char DH, Kroll SM, Miller T, *et al.* Irradiated uveal melanomas: cytopathologic correlation with prognosis. *Am J Ophthalmol* 1996;**122**:509–13.
4. Lommatzsch PK, Werschnik C, Schuster E. Long-term follow-up of Ru-106/Rh-106 brachytherapy for posterior uveal melanoma. *Graefes Arch Clin Exp Ophthalmol* 2000;**238**:129–37.
5. Gragoudas ES, Seddon JM, Egan K, *et al.* A. Long-term results of proton beam irradiated uveal melanomas. *Ophthalmology* 1987;**94**:349–53.
6. Harbour JW, Char DH, Kroll S, *et al.* Metastatic risk for distinct patterns of postirradiation local recurrence of posterior uveal melanoma. *Ophthalmology* 1997;**104**:1785–92; discussion 1792–3.
7. Gragoudas ES, Egan KM, Seddon JM, *et al.* Intraocular recurrence of uveal melanoma after proton beam irradiation. *Ophthalmology* 1992;**99**:760–6.
8. Lommatzsch P, Vollmar R. [A new way in the conservative therapy of intraocular tumors by means of beta-irradiation (Ruthenium 106) with preservation of vision] (In German). *Klin Monatsbl Augenheilkd* 1966;**148**:682–99.
9. Rouberol F, Roy P, Kodjikian L, *et al.* Survival, anatomic, and functional long-term results in choroidal and ciliary body melanoma after ruthenium brachytherapy (15 years' experience with beta-rays). *Am J Ophthalmol* 2004;**137**:893–900.
10. Damato B, Patel I, Campbell IR, *et al.* Visual acuity after Ruthenium(106) brachytherapy of choroidal melanomas. *Int J Radiat Oncol Biol Phys* 2005;**63**:392–400.
11. Kleinedam M, Guthoff R, Bentzen SM. Rates of local control, metastasis, and overall survival in patients with posterior uveal melanomas treated with ruthenium-106 plaques. *Radiother Oncol* 1993;**28**:148–56.
12. Jampol LM, Moy CS, Murray TG, *et al.* Collaborative Ocular Melanoma Study Group (COMS Group). The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: IV. Local treatment failure and enucleation in the first 5 years after brachytherapy. COMS report no. 19. *Ophthalmology* 2002;**109**:2197–206. Erratum in: *Ophthalmology* 2004 Aug;**111**(8):1514.
13. Melia M, Moy CS, Reynolds SM, *et al.* Collaborative Ocular Melanoma Study – Quality of Life Study Group. Quality of life after iodine 125 brachytherapy vs enucleation for choroidal melanoma: 5-year results from the Collaborative Ocular Melanoma Study: COMS QOLS Report No. 3. *Arch Ophthalmol* 2006;**124**:226–38.
14. Kreusel KM, Bechrakis N, Riese J, *et al.* Combined brachytherapy and transpupillary thermotherapy for large choroidal melanoma: tumor regression and early complications. *Graefes Arch Clin Exp Ophthalmol* 2006;**244**:1575–80.
15. Shields CL, Cater J, Shields JA, *et al.* Combined plaque radiotherapy and transpupillary thermotherapy for choroidal melanoma: tumor control and treatment complications in 270 consecutive patients. *Arch Ophthalmol* 2002;**120**:933–40.
16. Harbour JW, Meredith TA, Thompson PA, *et al.* Transpupillary thermotherapy versus plaque radiotherapy for suspected choroidal melanomas. *Ophthalmology* 2003;**110**:2207–14; discussion 2215.
17. Shields CL, Shields JA, Cater J, *et al.* Transpupillary thermotherapy for choroidal melanoma: tumor control and visual results in 100 consecutive cases. *Ophthalmology* 1998;**105**:581–90.
18. Sagoo MS, Shields CL, Mashayekhi A, *et al.* Plaque radiotherapy for juxtaepapillary choroidal melanoma overhanging the optic disc in 141 consecutive patients. *Arch Ophthalmol* 2008;**126**:1515–22.
19. Bartlema YM, Oosterhuis JA, Journée-De Korver JG, *et al.* Combined plaque radiotherapy and transpupillary thermotherapy in choroidal melanoma: 5 years' experience. *Br J Ophthalmol* 2003;**87**:1370–3.
20. Wilson MW, Hungerford JL. Comparison of episcleral plaque and proton beam radiation therapy for the treatment of choroidal melanoma. *Ophthalmology* 1999;**106**:1579–87.
21. Karlsson UL, Augsburger JJ, Shields JA, *et al.* Recurrence of posterior uveal melanoma after 60Co episcleral plaque therapy. *Ophthalmology* 1989;**96**:382–8.
22. Damato B, Patel I, Campbell IR, *et al.* Local tumor control after 106Ru brachytherapy of choroidal melanoma. *Int J Radiat Oncol Biol Phys* 2005;**63**:385–91.
23. Weidner N, Semple JP, Welch WR, *et al.* Tumor angiogenesis and metastasis – correlation in invasive breast carcinoma. *N Engl J Med* 1991;**324**:1–8.
24. Damms T, Schäfer H, Guthoff R, *et al.* Correlation of histological tumor vascularization and Doppler sonography in patients with malignant melanoma of the choroid. *Graefes Arch Clin Exp Ophthalmol* 1995;**233**:257–60.
25. Resch H, Garhöfer G, Schmetterer L, *et al.* Choroidal perfusion in eyes with untreated choroidal melanoma. *Acta Ophthalmol* 2008;**86**:404–7. Epub 2007 Nov 26.
26. Lischko AM, Seddon JM, Gragoudas ES, *et al.* Evaluation of prior primary malignancy as a determinant of uveal melanoma. A case–control study. *Ophthalmology* 1989;**96**:1716–21.
27. Gündüz K, Shields CL, Shields JA, *et al.* Plaque radiotherapy of uveal melanoma with predominant ciliary body involvement. *Arch Ophthalmol* 1999;**117**:170–7.
28. Oosterhuis JA, Journée-de Korver HG, *et al.* Transpupillary thermotherapy: results in 50 patients with choroidal melanoma. *Arch Ophthalmol* 1998;**116**:157–62.
29. Hungerford JL. Current trends in the treatment of ocular melanoma by radiotherapy. *Clin Experiment Ophthalmol* 2003;**31**:8–13.